Cryptic genomic variation in a panmictic squid: an ancient inversion supergene and genome-wide signatures of positive selection in *Illex illecebrosus*

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Abstract

Marine species with high dispersal capabilities often exhibit minimal geographic population structure while maintaining significant adaptive genomic variation. In this study, we employ a chromosome-level reference genome and whole-genome data from 633 northern shortfin squid, *Illex illecebrosus*, collected across the Northwest Atlantic, to elucidate genomic variation previously inaccessible with a divergent reference assembly. Our findings confirm nearly complete range-wide panmixia, yet reveal a singular major axis of genomic structure, a ∼19 Mb inversion on chromosome 2. Its derived arrangement segregates at a frequency of 0.626, is present throughout the species' range without a geographic cline, and achieves $F_{ST} = 0.37$ between alternative homozygotes despite a geographic $F_{ST} \approx 0$. Its evolutionary trajectory is inconsistent with neutral drift. The common derived arrangement retains 2.3 times less diversity than expected given its current frequency, and barrier-aware coalescent modeling dates the recombination barrier between the arrangements to approximately 800–850 ka, indicating a prolonged period near frequency $p \approx 0.22$–0.38 followed by a rapid rise to its present frequency that ended at least ∼100 ka ago. The inversion is gene-poor and lacks functional enrichment, yet contains thousands of near-fixed differences concentrated within the non-recombining block, suggesting selection on the arrangement as a long-lived supergene rather than on a single locally adaptive locus. Outside the inversion, demographic analyses indicate strong expansion (genome-wide Tajima's $D \approx -2$) to a current effective population size of approximately seven million. A stratified scan, corroborated across methods, for recent positive selection identified 627 candidate sweep windows, including 24 high-confidence regions independently supported by multiple methods. Genes associated with these sweeps show significant enrichment for neurodevelopment and photoreception, with additional candidates involving lineage-specific genes and Z-linked RNA-regulatory and hormone-signaling loci. In males, linkage disequilibrium decays at the same rate on the Z chromosome as on the autosomes, which, after accounting for a female mutation rate two to four times the male rate, indicates that female meiosis is achiasmate, as in octopuses. Collectively, these findings reveal two contrasting modes of adaptation in an otherwise panmictic marine species, a long-lived chromosomal polymorphism whose derived arrangement rose under selection to an intermediate frequency, and recent directional selection on sensory and neural functions. More broadly, they illustrate how a conspecific reference genome can uncover structural and adaptive variation that remains largely undetectable through geographic differentiation alone.

1 Introduction

Population genetic theory predicts that species with vast census sizes and high dispersal will show weak geographic differentiation, because migration homogenizes allele frequencies faster than drift can differentiate them, and a large effective size makes selection efficient while further slowing drift. There are examples of large-N_e species with no geographic structure in terrestrial systems (needs reference), but the pattern is less well characterized in marine species, where geographic datasets have largely been limited to mitochondrial genomes or a few nuclear loci (Waples, 1998; Hauser and Carvalho, 2008; Gagnaire et al., 2015). More broadly, genomic analysis of cephalopods has lagged behind that of many commercially important vertebrates due to their large, repeat-rich, and extensively reorganized genomes relative to other molluscs (Albertin et al., 2015, 2022; Schmidbaur et al., 2022; Coffing et al., 2025, 2026; Small et al., 2026). Until recently, these limitations were particularly pronounced for *Illex illecebrosus*, as no conspecific chromosome-level reference genome was available.

Pelagic cephalopods exemplify a Marine system where we might expect large Ne and low geographic differentiation. Ommastrephid squids, which are short-lived, semelparous, highly fecund, and capable of extensive dispersal during both early life stages and adulthood, generate large and often highly variable populations whose abundance is closely linked to oceanographic conditions (O'Dor and Dawe, 1998; Arkhipkin et al., 2015). The northern shortfin squid, *Illex illecebrosus*, exemplifies this life history. It is an annual semelparous broadcast spawner distributed over more than 2,500 km of the Northwest Atlantic shelf and slope, from the Mid-Atlantic Bight to Newfoundland and Labrador, where it supports major commercial fisheries in both U.S. and Canadian waters (O'Dor and Dawe, 1998; Arkhipkin et al., 2015; Baker et al., 2025). Recruitment and distribution are strongly influenced by transport in the Gulf Stream and associated shelf and slope circulation, while adults undertake extensive horizontal and diel vertical migrations. Collectively, high dispersal, extreme fecundity, and large population size predict minimal persistent geographic structure, while providing a substantial substrate for natural selection.

The inaugural comprehensive population genomic analysis of *Illex illecebrosus* established a crucial baseline. Utilizing whole-genome data from 540 specimens collected from over 40 locations, Baker et al. (2025) identified an absence of geographic genetic structure across the species' Northwest Atlantic range. This finding aligns with the hypothesis of extensive connectivity facilitated by early-life dispersal and adult migration, thereby supporting the classification of *Illex illecebrosus* as a single, highly interconnected stock. Nevertheless, due to the lack of a conspecific genome assembly, the reads were aligned to the chromosome-level genome of the distantly related diamondback squid, *Thysanoteuthis rhombus*, which diverged approximately 51 million years ago, resulting in a mean alignment rate of merely ∼46%. This divergence inherently limits inference to the conserved genome fraction that can be reliably mapped, thereby diminishing the ability to detect lineage-specific sequences, structural polymorphisms, and rapidly evolving loci. Reference divergence matters most for inversions. Detecting an inversion requires mapping both its breakpoints and the long differentiated haplotypes between them, and both are lost where reads map poorly, so a population studied against a distant reference can appear to lack inversions it actually carries. Consistent with this, Baker et al. did not find evidence of large structural variants and explicitly acknowledged reference divergence as a constraint on their detection.

Chromosomal inversions are a prime example of such variation. Inversions inhibit recombination between alternative arrangements, thereby maintaining combinations of alleles as segregating "supergenes" (Kirkpatrick, 2010; Schwander et al., 2014; Thompson and Jiggins, 2014; Wellenreuther and Bernatchez, 2018; Faria et al., 2019), and their evolutionary significance arises not only from their breakpoints but also from the long haplotypes they preserve. Numerous well-documented

instances exhibit noticeable phenotypic effects or occur in spatially structured populations, such as mimicry in *Heliconius* (Jay et al., 2021), alternative reproductive morphs in the ruff (Lamichhaney et al., 2016), colony organization in fire ants (Wang et al., 2013), and life-history variation in *Mimulus* (Lowry and Willis, 2010). Consequently, much of the inversion theory has concentrated on local adaptation with migration, where suppressed recombination safeguards locally advantageous allele combinations from gene flow (Kirkpatrick and Barton, 2006; Charlesworth and Barton, 2018). That explanation has the least scope in a large, weakly structured population, so an inversion maintained there at intermediate frequency without geographic structuring presents a distinct challenge. Its persistence suggests balancing mechanisms such as overdominance, associative overdominance induced by recessive deleterious load (Zhao and Charlesworth, 2016; Gilbert et al., 2020; Jay et al., 2021), negative frequency dependence, or temporally fluctuating selection (Wittmann et al., 2017; Rudman et al., 2022), and because large effective size slows drift, the history of such a polymorphism can extend over hundreds of thousands of generations. Differentiating these histories from neutral drift necessitates reconstructing the age and frequency history of the arrangements while explicitly considering the recombination barrier between them (Navarro et al., 2000; Guerrero et al., 2012; Berdan et al., 2023).

A chromosome-level *Illex illecebrosus* genome facilitates the investigation of selection beyond large structural polymorphisms. Cephalopods present a unique context for this inquiry because of their extreme life history, which is associated with remarkable sensory and neural complexity. Coleoid genomes demonstrate significant lineage-specific reorganization (Schmidbaur et al., 2022), extensive RNA-editing-based transcriptomic plasticity (Liscovitch-Brauer et al., 2017), and sophisticated nervous and visual systems (Albertin et al., 2015, 2022; Styfhals et al., 2022; Songco-Casey et al., 2022). In squids, vision adapts to substantial changes in depth and illumination during diel vertical migration, with photoreception extending beyond the camera eye to include dermal opsins and light-responsive chromatophore systems (Mäthger et al., 2010; Ramirez and Oakley, 2015; Kingston et al., 2015). In a large, highly connected population where geographic F_{ST} offers limited utility for detecting adaptation, recent positive selection is likely to manifest primarily through local distortions in diversity, linkage disequilibrium, and the site-frequency spectrum. Accurate interpretation of these signals necessitates first establishing the demographic and genealogical context in which they emerged.

In this study, we employed a chromosome-level annotated reference genome for *Illex illecebrosus*, recently made available (Small et al., 2026), to reanalyze range-wide whole-genome variation with a significantly enhanced genomic resolution. Our dataset included 633 individuals distributed across 11 fishery management divisions (Baker et al., 2025), along with *I. coindetii* (Genoscope, 2025) as an outgroup for divergence-based inference and *I. argentinus* as a recently diverged species. Initially, we assessed whether the range-wide panmixia reported by Baker et al. (2025) persisted when reads were aligned to a conspecific reference and investigated the emergence of previously inaccessible structural variations. This analysis identified a single ∼19 Mb inversion on chromosome 2 that segregated across the range, despite an otherwise largely unstructured genome. Subsequently, we tested as to whether this inversion represented a neutral polymorphism or a selected supergene by integrating arrangement-specific diversity and divergence with a barrier-aware coalescent model to reconstruct its age and historical frequency trajectories. Concurrently, we inferred population history from the site frequency spectrum and linkage disequilibrium, explicitly evaluating whether the data necessitated a sweepstakes-reproductive-success models (fecundity-with-high-mortality), wherein a small and variable fraction of adults contributes disproportionately to each generation (Hedgecock and Pudovkin, 2011; Eldon and Wakeley, 2006; Árnason et al., 2023) rather than Kingman coalescent, and scanned the remaining genome for recent positive selection using complementary methods (Kern and Schrider, 2018; Alachiotis and Pavlidis, 2018; DeGiorgio et al.,

2016). These analyses elucidated two distinct forms of genomic adaptation in an otherwise panmictic species, a long-lived chromosomal polymorphism under selection and a genome-wide set of recent sweeps concentrated on neurodevelopmental and photoreceptive functions. Collectively, these studies demonstrated how a conspecific genome transformed the apparent genomic landscape of a highly connected marine species, revealing structural and adaptive variations that were largely undetectable through geographic differentiation alone.

2 Results

2.1 Sample, sex, and the absence of geographic structure

We conducted an analysis of 633 *Illex illecebrosus* individuals from eleven NAFO management divisions, covering approximately 37–50°N (fig. 1, table 1). Details of each collection event, including source and depth, are provided in Supplementary Table S8. The read coverage on the Z chromosome categorizes the sample into 342 ZZ males and 291 ZO females, aligning with a ZZ/ZO system (Coffing et al., 2025; Small et al., 2026) (fig. 2). Autosomal analyses incorporate both sexes, while the Z chromosome, where males are diploid and females are haploid, is examined separately (section 2.5).

Consistent with range-wide panmixia reported by Baker et al. (2025), population structure is observed in a specific genomic region, although it is not geographically linked. A windowed principal-component analysis along chromosome 2 identifies the three-cluster structure within the inversion (among-cluster variance $\eta^2 = 0.748$ within the inversion compared to $\eta^2 = 0.003$ in the collinear sequence, Supplementary Fig. S3), and two genome-wide tests confirm its uniqueness. First, the chromosome-2 karyotype does not account for variation on any other chromosome (in a per-chromosome PCA colored by karyotype, $\eta^2 = 0.76$ on chr2 versus ≤ 0.01 on all 45 others). Second, independent of the chromosome-2 labels, a Gaussian-mixture screen of each chromosome's leading principal component supports more than one cluster only on chromosome 2 (three clusters, ΔBIC = 199, cluster separation 8.6σ) and on the Z chromosome (two clusters, reflecting the expected ZZ/ZO hemizygosity split), while every other autosome forms a single cloud of points (fig. 2). The residual dispersion on minor PCA axes does not correlate with division, latitude, or coverage (Supplementary Fig. S1), and a KING-robust relatedness screen (Manichaikul et al., 2010) identifies no duplicate, parent-offspring, or full-sib pairs, with only a single second-degree pair among the 633 individuals (maximum kinship 0.16), indicating effective unrelatedness. Given that a per-SNP differentiation scan is the most sensitive test for local adaptation in such a panmictic population, we compared per-SNP Hudson F_{ST} among divisions (Bhatia et al., 2013) to a permutation null, in which division labels were shuffled among individuals 20 times. This null keeps the genotype calls, missingness, allele-frequency spectrum and linkage of the data and removes only geography. Across the complete per-chromosome callsets of the 632 individuals from the ten divisions with more than one sample, we evaluated every biallelic SNP with a minor-allele frequency above 0.05 (20,117,984 SNPs on the 45 autosomes, Supplementary Methods), and the empirical differentiation remains entirely within the null expectation. No SNP surpasses the permutation maximum, and the number of SNPs above each upper null percentile lies within the range of the permutations (2,319 above the 99.99th percentile against $1{,}998 \pm 352$, permutation $p = 0.24$), including within the chromosome-2 inversion (8 against 13 expected), so the upper tail holds no more candidates than chance (Supplementary Figs. S2, S19). Nucleotide diversity and Tajima's D are consistent across latitude, and pairwise F_{ST} among divisions is approximately 0 throughout (fig. 6). Consequently, there is no geographic local adaptation anywhere in the genome, and the inversion itself is geographically non-clinal (fig. 1, Supplementary Fig. S6 and Table S2). Furthermore, arrangement

frequency does not correlate with capture depth, which ranges from 3 to 613 m across collection events (Supplementary Table S8). Among 539 individuals with recorded depth, the correlation between depth and the number of 2+ chromosomes carried is $\rho = -0.08$ ($p = 0.08$), and within the six divisions sampled at multiple depths, it is $\rho = +0.04$ (within-division permutation $p = 0.49$). A weak negative slope across collection events ($p = 0.04$) is entirely attributable to the two shallow, jig-caught Newfoundland (3 K) samples and disappears when they are excluded ($p = 0.23$), indicating that it cannot be disentangled from gear and location effects.

2.2 A large, selected inversion on chromosome 2

Identity, breakpoints, and karyotypes. Local PCA (Li and Ralph, 2019) and a Gaussian-mixture karyotype assignment (95.3% confident) resolve three genotype clusters on chromosome 2, the karyotypes of two arrangements that we call 2a and 2+ (2a/2a $n = 254$, 2a/2+ $n = 284$ and 2+/2+ $n = 95$, fig. 3a), whose differentiation is karyotypic rather than geographic, with $F_{ST} = 0.37$ between 2a/2a and 2+/2+ inside the inversion against a geographic $F_{ST} \approx 0$. Between-karyotype F_{ST} computed along the whole chromosome is ≈ 0 (median 0.006) across collinear sequence and jumps to a median of 0.34 (maximum 0.53) inside the inversion (Supplementary Fig. S20), so the karyotypes are one panmictic population everywhere except within the barrier. The three karyotypes co-occur throughout the range with no latitudinal cline in arrangement frequency (derived-arrangement frequency 0.54–0.71 across divisions, fig. 1, Supplementary Fig. S6 and Table S2). Two independent lines pin the breakpoints to chr2:60.54–79.50 Mb (a ∼18.96 Mb inversion), namely a step-change in arrangement differentiation and a strand flip in a whole-chromosome alignment to *I. coindetii* (Supplementary Fig. S11), and an extended block of linkage disequilibrium coincides with the same interval (Supplementary Fig. S4).

The reference individual carries both arrangements. We treat 2a, at a present frequency of 0.626, as the derived arrangement and 2+, at 0.374, as the ancestral arrangement. The reference allele cannot polarize the arrangements, because the individual from which the reference was assembled is heterozygous for the inversion. Its two haplotype assemblies differ by an inversion at the breakpoints inferred from the population data (60.54 and 79.53 Mb), the haplotype inverted relative to the primary assembly carrying 2a and the other, like the primary assembly, carrying 2+ (Supplementary Table S3). Whole-chromosome alignment of both haplotypes to *I. coindetii* shows that chromosome 2 is extensively rearranged between the species, with large blocks translocated and inverted along its length, and the inversion region is part of this complex structure. A 37-Mb segment of the *I. coindetii* chromosome aligns to 2+ as one contiguous run, inverted as a whole between chr2:44.4 and 79.5 Mb, so 2+ keeps the *I. coindetii* order within the segment. In 2a the right part of the run, the segregating inversion, is inverted a second time, which breaks the *I. coindetii* adjacency at the left breakpoint. Arrangement 2+ therefore differs from *I. coindetii* across this segment by one inversion and 2a by two, which places 2a as derived. The segregating inversion shares its right breakpoint with the older inverted block, so this region has rearranged repeatedly. Because the coalescent model fitted below is symmetric in the arrangement labels, the polarity changes the interpretation of the frequency history but not the fitted age or frequencies.

The inversion is non-neutral. The primary evidence is per-arrangement diversity (table 2, fig. 3c). Inside an inversion the two arrangements behave as partially isolated demes (Navarro et al., 2000; Guerrero et al., 2012), and without gene flux between them the neutral diversity of each scales with its frequency (Navarro et al., 2000; Peischl et al., 2013). The common derived arrangement is instead less diverse than its frequency predicts, with $\pi_I/\pi_S = 0.737 \pm 0.026$ against

an equilibrium expectation of $p/(1-p) = 1.67$ for a polymorphism held at 0.626, a 2.3-fold deficit that is the signature of an arrangement that was rarer for most of its history and rose recently. A diffusion argument that uses neither the mutation rate nor the accessibility mask excludes neutral drift as the cause of that rise (section 6). Over a 100-ky rise, the middle of the fitted range, at the N_e prevailing then ($2.2–4.9 \times 10^6$), the standard deviation of allele frequency under pure drift is 0.045–0.068, so the observed rise of 0.25–0.41 is 3.6–8.9 standard deviations of drift. A rise four times slower would be only 1.4–3.4 standard deviations away, but a coalescent profile of the rise duration excludes it, with ΔD of 5.3 at 400 ky and 9.4 at 600 ky against a calibrated cutoff of 2.3 (Supplementary Fig. S14). Drift alone could, by contrast, have carried a new arrangement to the earlier plateau. The expected time for a new neutral mutation to reach a frequency of 0.22–0.38, conditional on reaching it, is 0.48–0.88 N_e generations, or 0.26–0.48 million generations at the ancestral N_e of 5.5×10^5, well within the inferred lifetime of the inversion. The departure from neutrality is therefore the recent rise, which drift cannot produce at the N_e of the time.

Age and trajectory of the inversion. Estimating an inversion's age requires a simulation in which the two arrangements exchange ancestry everywhere, but are limited across the inversion to only instances of gene flux. Since, standard coalescent simulators do not provide this specific barrier aware event, we therefore fit the inversion's age and frequency trajectory with msinv (Small, 2025), a sequentially Markov coalescent simulator with recombination that enforces the cross-karyotype recombination barrier, tracks ancestral material, and outputs tree sequences (section 6). The data constrain the shape of the frequency trajectory $p(t)$ rather than any site-level mechanism, so we condition on a deterministic trajectory under within-arrangement neutrality and fit it by grid search to π_S/π_I, d_{xy}/π_S, and the per-arrangement singleton fraction of the site-frequency spectrum (fig. 4b, Supplementary Fig. S14, Supplementary Methods). The coalescent age was determined to be ∼800–850 ky. The reconstructed history is that the recombination barrier formed ∼800–850 ka, coincident with the onset of the population expansion (section 2.3), the derived arrangement was held at $p \approx 0.22–0.38$ for most of its history, and then rose sharply to 0.626 over 50–200 ky, in a rise that ended at least ∼100 ka ago, under selection of order $N_e s \sim 10^2–10^3$ ($s \approx 1.2–5 \times 10^{-4}$). The fits used a 29.7-kb simulated inversion, and the fitted statistics change by less than 5% when extrapolated from that length to the 19-Mb inversion (Supplementary Methods). These ranges are calibrated 95% sets from a parametric bootstrap that includes the variance of the frequency trajectory under drift, and in simulated histories across the whole grid they contain the true value in 95–99% of datasets (Supplementary Methods). The model places the derived arrangement at its plateau frequency when the barrier forms, so t_{inv} dates the recombination barrier and not a single-copy origin of the arrangement (see Discussion). The observed joint site-frequency spectrum between arrangements shows the same barrier directly, with near-fixed differences and off-diagonal density in the inversion and a collapse to the diagonal in collinear controls (Supplementary Fig. S5). Note that absolute age scales with the assumed mutation rate $\mu = 3 \times 10^{-9}$, so different values of the mutation rate will change the absolute age.

Inversion as a supergene. The barrier to recombination is shown directly by the block of linkage disequilibrium that spans exactly the inversion (Supplementary Fig. S4), by the arrangement differentiation, and by the joint site-frequency spectrum (fig. 3b, Supplementary Fig. S5). Concluding that it is not locally adaptive by geography, since there is no cline and geographic $F_{ST} \approx 0$, and the migration-selection scenario that explains most well-studied inversions (Kirkpatrick and Barton, 2006) cannot apply. It is also not a region of strong selective sweep, because the genome-wide scan below finds no cross-method-concordant sweep inside the inversion. The

RAiSD footprint statistic is broadly elevated and the classifier over-calls across the inversion, but the two never colocalize into focal concordance, as expected for selection on the arrangement as a whole rather than on a focal swept locus, and the concordance requirement filters these calls. Gene flux between arrangements is not detectable in mean d_{xy}, though a residual ∼40% depletion of near-fixed differences appears in the outermost megabase around the breakpoints. The rise ended at an intermediate frequency that has persisted for at least 100 ky, whereas directional selection of the fitted strength would carry an arrangement from 0.626 to 0.99 within 8–35 ky. The rise therefore either approached a balanced equilibrium near the present frequency or reflects selection that ceased when the rise ended. The mechanism of a balanced equilibrium, whether overdominance, associative overdominance, frequency dependence, or a moving optimum under fluctuating selection, cannot be resolved from these data alone. A shelf-spawning squid whose paralarvae ride the Gulf Stream may well experience temporally fluctuating selection on ∼100 ky glacial cycles, but this remains a hypothesis.

Gene content and the architecture of arrangement divergence. The inversion has no functionally significant enrichment. Its 89 gene models (4.7 per Mb, against 6.8 per Mb genome-wide) place it at the 15th percentile of autosomal windows of the same size (median 127 genes, interquartile range 100–151), and the same for annotated content, since 45 genes have a cross-species functional annotation and 29 carry gene-ontology terms, again the 15th percentile (median 49 GO-annotated genes per window). Their functional classes are diverse (lipid metabolism, DNA repair, transcription and splicing, signalling) with no coherent pathway (Supplementary Table S7). A gene-ontology test of its annotated genes against the genome returns no enriched terms at FDR<0.05, and a permutation null on a random gene sets of the same size produce more nominal hits than the inversion genes (86.6±60 against 69, permutation $p = 0.51$), while the largest nominal per-term enrichments (∼40-fold) rest on as few as two genes (Supplementary Fig. S21). The inversion thus appears as a functionally unremarkable slice of the genome, and likely not a selected gene set but instead a supergene maintained for the arrangement as a whole. What the arrangements do accumulate is divergence as a unit, with 3,513 near-fixed differences between 2a and 2+ (per-arrangement $|\Delta AF| \geq 0.9$) are spread across the inversion (Supplementary Fig. S5). These are overwhelmingly non-coding since coding sequence is only ∼0.2% of the region, but the exonic fixed differences are ∼2.3-fold over-represented and clustered. All 19 fall in just nine genes, led by a cytochrome P450 (*CYP4X1*) and an acidic-repeat protein (four each) and including a phospholipase A2, the ciliary gene *RPGRIP1L*, and a cysteine-rich venom-like protein. Among coding sites inside the inversion, the most differentiated between arrangements are depleted rather than enriched for nonsynonymous (zero-fold degenerate) positions. Of the top 1% of coding sites by between-arrangement F_{ST}, 45% are zero-fold, against 82% of all coding sites in the inversion (odds ratio 0.18, $p = 2 \times 10^{-34}$), and per-transcript divergence between arrangements is insignificant relative to collinear controls (5 of 95 transcripts exceed the control 95th percentile, as expected by chance, Supplementary Methods).

2.3 Demographic history shows an expanding population

The age and selection estimates above depend on the population's demography and on the standard Kingman coalescent. Genome-wide, *Illex illecebrosus* shows $\pi \approx 0.0093$, close to the 1.06% heterozygosity estimated from k-mers of the individual used for the assembly (Small et al., 2026), and Tajima's $D \approx -2$ in essentially every window (Tajima, 1989a,b), an unambiguous, genome-wide signature of expansion (table 3, fig. 4a, fig. 6). A moments fit to the folded SFS gives growth from an ancestral $N_e = 547{,}928$ to a present $N_e = 6{,}808{,}096$ over 769,519 generations (∼12-fold, with

generations equal to years in this annual species), and this history reproduces both the observed π and a full-SFS Tajima's $D = -1.92$ under forward simulation. Stairway Plot 2 and GONE2 concur on strong recent expansion to N_e in the 1.5–1.7 $\times 10^7$ range. Because sweepstakes reproduction would mimic expansion by inflating rare variants (Eldon and Wakeley, 2006; Hedgecock and Pudovkin, 2011; Eldon et al., 2015), we tested the coalescent class directly. The folded SFS is fit far better by a Kingman-plus-growth model than by any multiple-merger (Beta) coalescent (Schweinsberg, 2003) (shape deviation $L_1 = 0.036$ vs a best-α 0.108), and contemporary N_e estimates from the SFS and from linkage disequilibrium agree where both are inconsistent with a strong multiple-merger process. The expansion began ∼770 ka, i.e. at essentially the same time as the recombination barrier formed, which places the supergene's origin at the start of the demographic episode that shapes the whole genome.

2.4 Recombination, background selection, and protein adaptation

Three further features of the genome were resolved to accurately identify selective sweeps. Recombination was estimated with ReLERNN (Adrion et al., 2020), a deep-learning method that infers the rate from linkage disequilibrium. The estimated rate is ∼0.24 cM/Mb on the autosomes (median 0.241, Supplementary Fig. S12). The map is flat with respect to gene content, and the rate differs by only ∼2% between windows adjacent to exons and windows >100 kb from any exon, unchanged when π is held fixed (Supplementary Fig. S17a). For example, on chromosome 1 the 2-Mb rolling median spans just 0.22–0.27 cM/Mb, the pericentromeric flanks where π halves show no more than a 7% dip, and the centromere itself (a 2.5-Mb block that is 80% repeat) carries no windows over its central 1.1 Mb (Supplementary Fig. S22). This pattern is similar genome-wide. Across the 44 autosomes in the map the chromosome-scale rolling median varies by 13–67% of each chromosome's median, and no chromosome shows a peak more than 17% above it. Background selection is moderate and gene-density-driven. Nucleotide diversity falls with local coding density ($r = -0.31$, a 20–30% reduction in gene-dense windows) and rises 43% between exon-adjacent and exon-distant windows (Supplementary Fig. S17) (Charlesworth et al., 1993; McVicker et al., 2009; Charlesworth and Jensen, 2021). A model-based B-map could not be fit under this regime (the classical approximation overflows where the deleterious mutation rate exceeds the selection coefficient), so local coding fraction serves as the background-selection covariate that the sweep scan controls for explicitly. Protein adaptation over the divergence timescale is substantial. McDonald–Kreitman tests (McDonald and Kreitman, 1991; Messer and Petrov, 2013a) against *I. coindetii* across 20,999 genes give a genome-wide proportion of adaptive substitutions $\alpha \approx 0.34$–0.64, with the splicing factor *SF3B4* the single most significantly adaptive gene (Supplementary Fig. S10). Per-gene power is limited, so we treat the MK results as complementary to the recent-sweep scan, which detects a different timescale of selection. Comparing to *I. argentinus* we find that it is recently diverged (an order of magnitude fewer fixed coding differences than *I. coindetii*, Supplementary Table S3) yet carries near-identical genome-wide diversity ($\pi = 0.0092$ vs 0.0093, within 1%), so the two are large-N_e, expansion-shaped congeners of matched standing variation. We also found twenty-two blocks of ≥1 Mb on 17 chromosomes (129 Mb, 3.9% of the autosomes) that seemingly carry 2–6 times the chromosome-median π with less negative Tajima's D (Supplementary Table S9). They are present at the same strength in every sampling location, are no more differentiated among locations than flanking sequence, and lack the long-range linkage disequilibrium of a segregating inversion (Supplementary Figs. S23, S24). Read-depth analyses resolve them into three categories. In six, depth is raised in every individual (1.2–2.2 times the flanks across the population sample, and up to 2.7 and 2.4 times in long reads from the individual used for the assembly and a second individual (Small et al., 2026)), and a single diploid genome carries five to six times the flanking

density of heterozygous sites at skewed allele fractions. We interpret this as a duplicated sequence collapsed into one copy in the assembly. In eight, short-read depth is reduced to 0.6–0.9 times the flanks in every individual. A third of their accessible sequence is not uniquely mappable by 150-bp reads, against 4% genome-wide, so these blocks are largely duplicated within the assembly itself and short reads there map to several places and are lost at the mapping-quality filters. Long reads, which map uniquely through the duplicated sequence, show a much smaller deficit, and divergent haplotypes may add to the loss. The remaining eight have normal depth and mapping and are gene-poor regions of genuinely elevated diversity. Since our summary statistics cannot separate any of these from the footprints of selection, all 22 blocks were excluded from the sweep scan and the gene-set analyses.

2.5 A genome-wide scan for selective sweeps

We scanned for recent positive selection with diploS/HIC (Schrider and Kern, 2016; Kern and Schrider, 2018), a five-class site-frequency/diversity-based classifier (neutral, hard, soft, linked-hard, and linked-soft, in 1.1 Mb feature windows of eleven 100 kb sub-windows) trained on forward and coalescent simulations under the fitted growth demography with the final model reaching 81% class accuracy (fig. 5, Supplementary Figs. S7 and S8). Applied to the genome, the classifier over-calls sweeps, assigning only 1% of windows to the neutral class and 91% to the linked-hard or linked-soft classes. We determined this was because real neutral windows (accessible, $\geq$5 kb from any gene) carry 3–8-fold more within-window spatial heterogeneity in π and θ than the simulations, even though their marginal SFS matches ($D \approx -2$). We went further and simulated our identical missing-data structure and sample incompleteness ultimately ruling that out as the cause. The residual is real per-sub-window diversity variation (mutation-rate variation plus background selection) that a uniform-θ coalescent does not reproduce (Johri et al., 2022). We therefore use the classifier as a stratified empirical-outlier detector, in which the sweep score ($p_{\text{hard}} + p_{\text{soft}}$) is ranked within strata of callable fraction and coding density, so a window is an outlier only relative to windows of similar accessibility and gene density, and calls are corroborated by an independent LD-footprint method, RAiSD (Alachiotis and Pavlidis, 2018), or a composite-likelihood SFS method, SweepFinder2 (Nielsen et al., 2005; DeGiorgio et al., 2016). Thus, we report sweeps at two levels of stringency (Supplementary Fig. S13). Tier 1 is the diploS/HIC hard/soft classification, spatially resolved by a recombination-aware Markov decoder, a hidden Markov model over the per-window (100 kb) classes with neutral, linked, and sweep-centre states whose emissions are the prior-corrected class posteriors and whose dwell lengths follow the $\sim$400 kb sweep footprint scaled by the recombination map (Supplementary Fig. S13a). Because the genome reads as $\sim$91% linked-selected under strong expansion, the decoder's role is spatial coherence. It retains hard/soft calls that lie in contiguous sweep centers and discards spatially isolated ones, reducing 1,651 raw calls to 677 windows (623 localized autosomal calls, 437 hard and 186 soft) and preferentially keeping the higher-outlier windows (median empirical-outlier percentile 0.98 vs 0.95 for discarded calls, Supplementary Fig. S13b). Tier 2 is the subset independently corroborated by a other methods, producing 24 concordant regions (a diploS/HIC-HMM outlier plus RAiSD or SweepFinder2). The two tiers agree strongly, with 88% of windows at concordant loci decoded as sweep centers versus 2.9% genome-wide (odds ratio 242, $p < 10^{-30}$), and 21 of the 24 concordant regions contain a Tier-1 center (Supplementary Fig. S13c), so the conservative concordance set is a high-purity subset of the classifier calls, not a disjoint list. The Z chromosome is necessarily diploS/HIC-only and scanned separately in males (below), it contributes four further Tier-1 calls, for a genome-wide Tier-1 total of 627 (fig. 5, Supplementary Fig. S18).

Neurodevelopment and photoreception. The 623 autosomal Tier-1 calls overlap 885 genes (475 with GO annotation). One-third of calls fall in gene-poor, often transposon-rich windows and are excluded from enrichment analyses. A gene-ontology enrichment against the annotated-gene background (Fisher's exact test with Benjamini–Hochberg FDR, Benjamini and Hochberg 1995) returns 124 terms at FDR<0.05 (Supplementary Fig. S15, Table S4), dominated by neural development (neurogenesis, generation of neurons, neuron development, neuron-projection and cell morphogenesis, at 1.5–1.8-fold), together with transcriptional and RNA-regulatory control, and a weaker light-response signal (response to light stimulus 1.9-fold, response to radiation 1.8-fold). The enrichment is carried by the hard-sweep calls (98 terms), and the disjoint soft-sweep set has none at FDR<0.05. We further tested there regions aginst a hypothesis of background selection, since gene-dense regions have lower diversity. We therefore screened every Tier-1 call with the same per-region background-selection test used for the concordant set, retaining only calls whose local π falls below the background-selection expectation given flanking coding density. This keeps 291 of the 623 (237 hard, 54 soft). The BGS-robust set yields 151 terms at FDR<0.05, and every leading term's fold enrichment rises over the previous set, including response to light stimulus $1.94 \rightarrow 3.15$ (FDR $2 \times 10^{-2} \rightarrow 9 \times 10^{-5}$), response to radiation $1.76 \rightarrow 2.79$ (FDR $3 \times 10^{-2} \rightarrow 2 \times 10^{-4}$), neuron-projection morphogenesis $1.81 \rightarrow 2.25$, cell morphogenesis $1.73 \rightarrow 2.24$, neuron development $1.59 \rightarrow 1.97$, generation of neurons $1.59 \rightarrow 1.90$, transcription by RNA polymerase II $1.73 \rightarrow 2.08$ (Supplementary Fig. S15). The photoreception signal, the weakest in the full set, becomes the single most enriched theme after the BGS screen, and its more specific child terms (detection of light stimulus $2.5 \rightarrow 3.9$, phototransduction $2.5 \rightarrow 3.4$) move in the same direction, though with too few genes to reach significance. A shared-gene enrichment map of the BGS-robust set resolves the same two connected modules, neural development/morphogenesis and light response, with no third (Supplementary Fig. S16, Table S5).

Candidate sweeps. The Tier-2 concordant regions carry a biologically coherent set of cephalopod-orthologous genes that sit squarely within these themes (table 4, Supplementary Table S6). They include the neural/EMT transcription factor *ZEB2*, the synaptic-vesicle GTPase *Rab3*, the apelin receptor *Aplnr* (cardiovascular/vasculature signalling), the DNA-damage-response gene *MACROD2*, the phospholipid synthase *cept1*, and the ceramide phosphoethanolamine synthase *Cpes*. A per-region background-selection test isolates 11 of these as high-confidence candidates in which diversity is reduced beyond the background-selection expectation, and fine-scale localization confirms all 11 as recent sweeps by the sign of Tajima's D at the diversity valley. It reassigns two apparent gene-desert signals to transposable-element regions (flagged lower confidence) and moves two targets to neighboring genes, from *eif3h* to *Cpes* at chr1:85.5 Mb and to *GALT* at chr25:65.0 Mb. The single most distinctive autosomal candidate is a cephalopod-lineage-specific gene under a hard sweep on chromosome 16, a tandem-exon transcript with no functional annotation whose translation matches an uncharacterised protein family in the bobtail squid *Euprymna* ($E = 2 \times 10^{-53}$) (Albertin et al., 2015, 2022). There is no overlap between the sweep candidates and the MK-adaptive genes.

The Z chromosome. Because males are Z-diploid and females Z-haploid, the Z was scanned separately in the 330 confident ZZ males, using a model retrained at the matched sample size and accounting for the Z's reduced effective size. The male-only screen was validated against the 289 haploid females, whose per-window diversity tracks the males' at $r = 0.996$. Of the four Z Tier-1 calls (three hard and one soft, Supplementary Fig. S18), three credible, gene-localised sweeps survive fine-scale triage, all confirmed as conserved cephalopod genes by direct homology. These are *Pasilla*/Nova-1 (a neuronal alternative-splicing regulator with 92% identity to its *Octopus* ortholog),

Smaug/samd4a (a germline RNA-binding translational repressor), and *Esrrg* (an estrogen-related receptor with an adjacent likely *cis*-regulatory lncRNA, with 97.5% identity to its *Octopus* ortholog), a coherent RNA-regulation-and-hormone-signalling signature on a sex chromosome that is dosage compensated, enriched for chromatin-modifying genes and carries two conserved sex-biased lncRNAs with RNA-binding-protein sites (Small et al., 2026). *Amnionless*, with the female-biased lncRNA *Zfest* transcribed antisense to it (Small et al., 2026), falls just outside the swept valley and shows only purifying selection, so it is a linked passenger rather than the target.

3 Discussion

The inversion is a datable supergene whose derived arrangement rose under selection. Chromosomal inversions are often maintained by local adaptation in the presence of gene flow, producing geographic clines in arrangement frequency (Kirkpatrick and Barton, 2006; Lowry and Willis, 2010; Faria et al., 2019). That explanation does not fit *Illex illecebrosus*. The chromosome-2 inversion is distributed throughout the species' range, its frequency is non-clinal, and geographic differentiation is effectively absent. Instead, the inversion appears to represent a long-lived polymorphism under selection at the level of the arrangement. The derived arrangement (2a) retains substantially less diversity than expected from its current frequency, indicating that it was at lower frequency in the past. A mutation-rate-independent drift argument, together with a barrier-aware coalescent model, supports an age of approximately 800,000–850,000 years and a history in which the derived arrangement remained near a frequency of 0.22–0.38 for much of its lifetime before rising to 0.63 in a rapid rise that ended at least 100,000 years ago, and it has held at that intermediate frequency since.

This history differs from that of classic *Drosophila* inversions, whose frequencies often vary across geographic or seasonal gradients (Corbett-Detig and Hartl, 2012; Kapun and Flatt, 2019), and from the human 17q21 inversion, whose increase is more recent (Stefansson et al., 2005), although the *Illex illecebrosus* history also ends in a selected rise. Why that rise stopped at an intermediate frequency is the open question. One possible analogue is the mimicry supergene of *Heliconius numata*, in which an inverted arrangement carries recessive deleterious variation that contributes to associative overdominance and limits its frequency (Jay et al., 2021). Associative overdominance is therefore one plausible mechanism in *Illex illecebrosus*, particularly because large non-recombining regions can accumulate recessive load (Zhao and Charlesworth, 2016; Gilbert et al., 2020; Berdan et al., 2021). The inversion contains clustered coding divergence, including 19 fixed exonic differences across nine genes, providing potential substrate for such a load, although the differentiated coding sites are not enriched for nonsynonymous changes. Other mechanisms remain possible. A prolonged plateau followed by a rise could also arise under a changing selective optimum (Matuszewski et al., 2015), and temporally fluctuating selection can maintain polymorphism over long periods in natural populations (Wittmann et al., 2017; Rudman et al., 2022). For a shelf-spawning squid whose paralarvae are transported by the Gulf Stream, repeated environmental change across glacial cycles provides a plausible ecological context.

The data therefore resolve the historical trajectory more clearly than the selective mechanism. They support selection acting on the inversion as an integrated genomic unit rather than on a single identifiable locus within it. The region is gene-poor, lacks functional enrichment, contains thousands of highly differentiated variants distributed across the block, and shows no evidence of a localized selective sweep. Together, these features are consistent with long-term selection on the alternative chromosomal arrangements as supergenes (Kirkpatrick, 2010; Berdan et al., 2023).

The derived arrangement may not have started from a single copy. The fitted history places the derived arrangement at 0.22–0.38 when the recombination barrier forms, so the barrier age is the age of a polymorphism already at intermediate frequency and not necessarily that of a single new inversion. A start from more than one copy is plausible here for structural reasons. Chromosome 2 is extensively rearranged between *Illex illecebrosus* and *I. coindetii*, and the segregating inversion lies inside an older inverted block whose right breakpoint it reuses. This part of chromosome 2 has therefore rearranged repeatedly, and a region whose breakpoints are reused may also produce the same arrangement more than once, giving the derived arrangement several independent founding copies. The polarity rests on two outgroups, and chromosome 2 is rearranged in this region in both. *I. coindetii* keeps the 2+ order only across the left breakpoint, because the older inverted block shares the right breakpoint. Chromosome 2 of the more distant *Sthenoteuthis oualaniensis* (Li et al., 2022), a member of the same family, is rearranged throughout relative to *Illex illecebrosus*, but both inversion breakpoints fall inside collinear blocks. At each breakpoint it keeps the 2+ adjacency and lacks the 2a adjacency (Supplementary Table S3), so the two outgroups agree that 2+ is ancestral at both breakpoints. Because chromosome 2 rearranges so often in this lineage, the common ancestor may still have carried an intermediate form elsewhere in the region, and this agreement is support rather than proof. At most sites that differ between the arrangements 2+ carries the *I. coindetii* allele, and the minority at which 2a carries it fit variation that was segregating when 2a formed and was captured by it, possibly in more than one founding event (Supplementary Methods). The fitted age and plateau do not depend on this choice, because the model is symmetric in the arrangement labels. Other routes lead to the same soft start. The arrangement may, for example, have reached intermediate frequency in a partially isolated population and entered the main population by secondary contact. These alternatives and a single-copy origin predict different genealogies for the derived arrangement near its origin, and distinguishing them would require comparing a single-copy origin directly with a start at intermediate frequency.

Methodological lessons. Two methodological lessons from this analysis may be broadly useful for inversion studies. First, estimates of inversion age can be based on divergence between the two arrangements, which is symmetric with respect to their ancestral and derived states. The age estimate therefore does not depend on polarization, which matters here because the reference is uninformative and the outgroup comparison is not simple. The *Illex illecebrosus* reference individual is heterozygous for the inversion, so the reference orientation carries no information about ancestry, and the outgroup comparison rests on a region that has rearranged repeatedly. More generally, assuming that the reference orientation is ancestral creates a circularity in inversion polarization. Phased haplotype assemblies of the reference individual and the order of outgroup sequence across the whole rearranged segment, not only its orientation, polarize an inversion independently of the reference sequence.

Second, the genealogy of a segregating inversion is not adequately represented by treating the alternative arrangements as freely recombining chromosomes or as allelic states at a single locus. Suppressed recombination between arrangements instead creates a structured coalescent resembling two partially isolated demes separated by a recombination barrier (Navarro et al., 2000; Guerrero et al., 2012). Modeling this history therefore requires the barrier to be represented explicitly in the ancestral recombination graph. The `msinv` simulator implements this process, allowing the age and frequency history of the inversion to be fitted jointly from within-arrangement diversity, divergence between arrangements, and the per-arrangement site-frequency spectrum. In *Illex illecebrosus*, this framework resolves not only the age of the inversion but also its prolonged intermediate-frequency phase and subsequent rise, features that would be lost by reducing the inversion to a single-locus

polymorphism. As genomic surveys identify increasing numbers of segregating inversions in natural populations (Wellenreuther and Bernatchez, 2018; Mérot et al., 2020), reconstructing their age and frequency trajectories may provide a useful way to distinguish among inversion histories even when their phenotypic effects are unknown.

Selective sweeps are enriched near neural and photoreceptive genes and are robust to background selection. The genome-wide sweep scan identified two prominent functional themes. The first encompasses neurogenesis, neuron development, and neuron-projection and cell morphogenesis, consistent with a coleoid lineage characterized by a large, centralized, and cell-type-rich nervous system (Albertin et al., 2015; Styfhals et al., 2022). Coleoid genomes also show extensive regulatory reorganization around neural genes (Schmidbaur et al., 2022). A related signal has been reported in another coleoid lineage: in the sibling octopuses *Octopus bimaculatus* and *O. bimaculoides*, genes associated with brain and eye development are enriched among targets of positive selection (Coffing et al., 2026). Together, these observations suggest that neural and visual functions may be recurrent targets of selection across independently evolving coleoid lineages.

The second theme involves photoreception and response to light. Cephalopod photoreception extends beyond the camera eye, with opsins and components of phototransduction pathways expressed in the skin and chromatophore system (Ramirez and Oakley, 2015; Kingston et al., 2015). Their visual systems are also unusually complex, including highly differentiated optic lobes that evolved independently of the vertebrate visual system (Songco-Casey et al., 2022; Styfhals et al., 2022; Yoshida et al., 2015). In *Illex illecebrosus*, this signal has a plausible ecological context. The species is a visually guided predator that undergoes diel vertical migration through strong gradients in illumination and large seasonal movements across latitude (O'Dor and Dawe, 1998). These observations do not identify the selective pressure responsible for individual sweeps, but they provide a biological context for the repeated association of sweep candidates with light-response genes.

A central concern is whether these enrichments could arise from background selection rather than recent positive selection. Diversity is reduced in gene-dense regions of the *Illex illecebrosus* genome, as expected under background selection (Charlesworth et al., 1993; McVicker et al., 2009; Charlesworth and Jensen, 2021), and neural-development genes could therefore be over-represented among low-diversity windows even in the absence of sweeps. We addressed this directly by requiring candidate regions to show a reduction in diversity beyond that expected from local coding density. Rather than weakening after this filter, the major functional signals became stronger: enrichment for response to light increased from 1.9- to 3.2-fold, response to radiation from 1.8- to 2.8-fold, and morphogenesis-related terms from approximately 1.7–1.8- to 2.2–2.3-fold. Light response, initially the weaker of the two major themes, became the strongest after the background-selection screen. Thus, the neural and photoreceptive enrichments cannot be readily explained by gene density and its associated reduction in neutral diversity alone.

We nevertheless treat the individual sweep loci as candidates rather than confirmed targets of positive selection. Simulation-trained window classifiers such as diploS/HIC provide a powerful framework for detecting recent selection (Schrider and Kern, 2016; Kern and Schrider, 2018), but their absolute classification proved poorly calibrated in this genome. Although the simulated data reproduced the fitted demography, empirical sample size, and missing-data structure, putatively neutral genomic windows contained substantially more fine-scale heterogeneity in diversity than the simulations. This mismatch caused widespread classification of real genomic windows as linked to sweeps. We therefore did not interpret the raw classifier output directly. Instead, we ranked sweep scores empirically within strata of callable sequence and coding density, required spatial coherence with a recombination-aware decoder, and used independent RAiSD or SweepFinder2 support to

define the most conservative candidate set. An additional background-selection screen then tested whether local diversity was lower than expected from coding density. This layered approach follows the broader principle that selection scans should explicitly accommodate genomic features that cannot be reproduced adequately by the null model (Johri et al., 2022).

Hard-sweep classifications predominate among the retained candidates, which is potentially interesting because large populations with abundant standing variation are often expected to provide favorable conditions for soft sweeps (Messer and Petrov, 2013b). Hard sweeps can instead arise when adaptation depends on new or initially rare variants, particularly when the mutational target is limited (Hermisson and Pennings, 2005; Harris et al., 2018). However, we do not regard the apparent excess of hard sweeps as strong evidence for a particular mode of adaptation. The soft-sweep class is less well resolved in these unphased data, and the imperfect calibration of the classifier further limits inference about the relative prevalence of hard and soft sweeps. The more robust conclusion is therefore the localization and functional enrichment of candidate sweep regions, rather than their assignment to a specific sweep class.

Demography, sweepstakes reproduction, and the interpretation of selection. The demographic history of *Illex illecebrosus* is important not only as a description of population history, but also because it determines the genealogical null against which both the inversion and the sweep scan are interpreted. Genome-wide Tajima's D is approximately -2 across nearly all of the accessible genome, and the folded SFS is consistent with a major historical expansion. The fitted moments model places the onset of this expansion at approximately 770,000 years ago, with N_e increasing from about 5.5×10^5 to 6.8×10^6, a roughly 12-fold increase. Stairway Plot 2 and GONE2 recover the same qualitative history and similarly large contemporary effective population sizes. For an annual, highly fecund pelagic species, this combination of a large N_e, abundant rare variation, and very weak geographic differentiation is consistent with a population that has remained both large and highly connected over much of its recent evolutionary history.

High fecundity alone, however, makes this interpretation non-trivial. Marine broadcast spawners are canonical candidates for sweepstakes reproductive success, in which a small and variable fraction of adults contributes disproportionately to subsequent generations (Eldon and Wakeley, 2006; Hedgecock and Pudovkin, 2011). Under sufficiently extreme reproductive skew, genealogies can depart from the binary-merger Kingman coalescent and instead approach a multiple-merger coalescent, producing an excess of rare variants that can resemble demographic expansion. Recurrent selective sweeps can generate related distortions; in Atlantic cod, pervasive linked selection has been proposed to contribute to multiple-merger-like genealogies (Árnason et al., 2023). Thus, the strongly negative Tajima's D in *Illex illecebrosus* cannot by itself distinguish population growth from reproductive or selective processes that reshape genealogies.

We therefore tested the coalescent class directly rather than assuming that the life history of *Illex illecebrosus* implied a multiple-merger process. The folded SFS is substantially better fit by Kingman coalescence with population growth than by any of the Beta-coalescent models examined, and contemporary N_e inferred independently from the SFS and linkage disequilibrium is mutually consistent under the growth model. These results do not imply that reproductive success is equal among individuals, nor do they exclude sweepstakes reproduction in individual cohorts. Rather, they indicate that reproductive skew is not sufficiently strong or persistent to dominate the genome-wide genealogical process at the temporal and genomic scales represented by these data. Despite its extreme fecundity and semelparous life history, *Illex illecebrosus* therefore does not show the genome-wide multiple-merger signature that might otherwise be expected for a marine broadcast spawner.

This distinction has direct consequences for the evolutionary inferences in this study. First, it supports the use of a Kingman-based demographic history for estimating the age and frequency trajectory of the chromosome-2 inversion. Second, it provides a demographic null for the sweep scan in which the genome-wide excess of rare variants is attributed primarily to population growth rather than being treated as evidence of pervasive positive selection. The strong expansion also helps explain why raw sweep classifiers trained under simpler neutral models performed poorly: the baseline genome is already strongly displaced toward rare variants, leaving recent selection to be identified as a local departure from an unusually non-equilibrium genomic background. Establishing the coalescent and demographic context is therefore a prerequisite for interpreting both forms of selection described here.

The timing of the expansion also places the chromosome-2 inversion in a particularly informative historical context. The inferred age of the recombination barrier, approximately 800,000–850,000 years ago, overlaps the estimated onset of population expansion at approximately 770,000 years ago. The barrier between the arrangements therefore appears to have formed near the beginning of the demographic episode that generated much of the standing variation observed today. Yet its subsequent history differs sharply from the genome-wide demographic trajectory. While effective population size increased by roughly an order of magnitude, the derived arrangement remained near a frequency of $p \approx 0.22$–0.38 for much of its history before rising to its present frequency of 0.626. Population growth alone therefore cannot explain the inversion trajectory. The arrangement held its frequency across a profound change in population size and rose only much later. This temporal separation reinforces the inference that the inversion's history reflects selection acting on the arrangement rather than a passive response to the demographic expansion.

Linkage on the Z chromosome indicates achiasmate female meiosis. Sex-specific recombination has not been measured directly in any cephalopod (Coffing et al., 2026), but the ZZ/ZO system (Coffing et al., 2025; Small et al., 2026) provides an indirect test. The Z recombines only in ZZ males and spends two thirds of its time in them, so its population-scaled recombination rate relative to the autosomes depends on the female rate. With a Z effective size of three quarters of the autosomal value, equal recombination in the two sexes predicts a Z-to-autosome ratio of 0.5, whereas achiasmate female meiosis predicts 1 (Coffing et al., 2026). In males, linkage disequilibrium decays at the same rate on the Z as on the autosomes ($\rho_Z/\rho_A = 1.04$, 95% CI 0.88–1.28). Interpreting this ratio requires the effective size of the Z, because Z diversity is lower than effective size alone would predict. Synonymous divergence from *S. oualaniensis* shows that the Z mutates more slowly than the autosomes, with a female mutation rate two to four times the male rate ($\alpha = 0.33$, 95% CI 0.10–0.70), and the female bias persists after matching genes on selective constraint and base composition. The remaining reduction in Z diversity corresponds to a Z effective size of 0.68–0.76 of the autosomal value, close to the expectation under an equal breeding sex ratio, and together these estimates place the female recombination rate near zero (Supplementary Table S10). The same conclusion was reached from the Z of the two-spot octopuses, where a female-biased mutation rate was also found (Coffing et al., 2025, 2026). Because the Z has been conserved as a single syntenic unit across coleoids for more than 300 million years (Coffing et al., 2025; Small et al., 2026), achiasmate female meiosis and female-biased mutation appear to be shared features of the coleoid ZZ/ZO system, consistent with the reduction or loss of recombination in the heterogametic sex that is widespread across animals (Burt et al., 1991; Lenormand and Dutheil, 2005).

4 Limitations

Several uncertainties limit the resolution, rather than the overall interpretation, of our results. Most importantly, the callset is unphased, preventing the use of haplotype-homozygosity statistics and reducing our ability to distinguish hard from soft sweeps. The simulation-trained classifier also remains imperfectly calibrated to the empirical genome because real genomic windows contain substantially more fine-scale diversity heterogeneity than the simulated null. Our empirical stratification, spatial filtering, cross-method corroboration, and background-selection screen reduce sensitivity to this mismatch but do not provide an absolute false-positive rate. We therefore place greater confidence in the recurrent functional enrichment and in candidates supported by multiple lines of evidence than in the classification of individual loci or their assignment to a particular sweep mode. The inversion analysis has different limitations. Its absolute age depends on the assumed mutation rate. The fitted history imposes selection only through a trajectory of the arrangement frequency, deterministic in each simulation with its variance under drift added to the fit, and treats variants within each arrangement as neutral, which is the model our conclusion implies, with selection acting on the arrangement as a unit rather than on individual sites within it. The data therefore constrain when the arrangements diverged and how their relative frequencies changed, but not the selective mechanism responsible for their long-term maintenance. The drift bounds also assume a panmictic history. Structured neutral histories cannot be excluded in principle, and the most obvious alternative is an allopatric origin followed by secondary contact. In an isolate of effective size N_B, a new neutral inversion takes about 0.5–0.9 N_B generations to reach the inferred plateau of 0.22–0.38, and the isolation cannot have lasted much longer than the ∼800–850-ky coalescent age of the arrangements, so drift within an isolate even as large as the ancestral population ($\sim 5 \times 10^5$) could have carried the inversion to its plateau. A small isolate would also leave the derived arrangement less diverse than the ancestral one, as observed. An allopatric origin therefore remains possible, but it does not remove the need for selection, because it does not explain the recent rise, which drift cannot produce in the large panmictic population of that time. In particular, the present analyses cannot distinguish among associative overdominance, frequency-dependent selection, or temporally varying selection, nor can they identify a phenotype associated with the alternative arrangements. Comparative sampling of the inversion in closely related *Illex* species and direct phenotypic comparison among karyotypes would provide important tests of these alternatives.

Finally, functional annotation remains imperfect. Functional interpretation relies heavily on homology to relatively distant cephalopod models, leaving many lineage-specific genes poorly characterized. The neural and photoreceptive signals should therefore be interpreted primarily as gene-set-level enrichments, while individual candidate genes remain hypotheses for functional validation.

5 Conclusions

The population genome of *Illex illecebrosus* is strikingly homogeneous geographically but not evolutionarily. A conspecific chromosome-level reference reveals genomic variation that was largely inaccessible when population structure was viewed through a divergent reference and geographic differentiation alone. Most notably, a large chromosome-2 inversion persists across an otherwise panmictic population as a deeply differentiated, long-lived polymorphism whose history is inconsistent with neutral drift. Its lack of a geographic cline or obvious functional specialization makes it distinct from many classic inversion systems and emphasizes that the evolutionary history of a supergene can sometimes be reconstructed more clearly than its phenotype or selective mechanism.

Beyond the inversion, the genome records a different mode of selection in localized candidate sweep regions enriched near genes involved in neural development and photoreception. Together, these results show that extensive dispersal and negligible geographic structure do not imply an absence of adaptive genomic variation. Instead, selection in a highly connected marine population can be expressed through persistent structural polymorphism and localized genomic change without producing conventional population subdivision. More broadly, the study illustrates how appropriate reference genomes and analyses that explicitly account for demography, recombination, and genomic heterogeneity can expose evolutionary variation that otherwise remains cryptic.

6 Methods

The main steps are summarised here. Parameter values, thresholds, and software versions are given in the Supplementary Methods and Supplementary Table S1. The scripts for every step are in the `pipeline/` directory of the manuscript repository, and the inversion analysis is in the msinv repository (Data and code availability).

Samples and variant calling. We analysed all 633 *Illex illecebrosus* individuals whole-genome sequenced by Baker et al. (2025) (ENA project PRJEB101095), collected in 2022 and 2023 across eleven NAFO divisions, with coordinates, date, and capture depth recorded for each collection event (Supplementary Table S8). That study analyzed 540 of them after removing individuals with more than 25% missing genotypes, a consequence of mapping to the distantly related *Thysanoteuthis* reference. We re-processed the raw reads from scratch against the conspecific `Illex_F24` primary assembly (Small et al., 2026) (4.14 Gb, 45 numbered chromosomes plus the Z), and against this reference no individual needed to be excluded. Reads were trimmed with fastp (Chen et al., 2018), mapped with BWA-MEM2 (Vasimuddin et al., 2019), and deduplicated with Picard, and genotypes were called jointly across individuals with GATK HaplotypeCaller in gVCF mode (Poplin et al., 2018), all within grenepipe (Czech and Exposito-Alonso, 2022). The joint callset retains invariant sites. Variants were filtered on site quality, mapping quality, strand bias, missingness, and excess heterozygosity at intermediate frequency, and sites in GC-corrected high-depth windows or in annotated repeats were excluded, leaving 1.43 Gb of accessible sequence, of which 96% is uniquely mappable by 150-bp reads according to GenMap (Pockrandt et al., 2020). For the sweep scan and gene-set analyses this accessibility mask was extended by the 22 anomalous-diversity blocks described in the Results (Supplementary Table S9). Genotypes are unphased.

Diversity, site-frequency spectra, and demography. Mean sequencing depth is about 9-fold, so diversity statistics were estimated from genotype likelihoods rather than from called genotypes. Folded site-frequency spectra, π, Watterson's θ, and Tajima's D were computed with ANGSD (Korneliussen et al., 2014) directly from the BAM alignments rather than from the VCF, genome-wide, per chromosome, per NAFO division, and per karyotype, with the Z analyzed in males only. Demographic history was inferred with Stairway Plot 2 (Liu and Fu, 2020), moments (Jouganous et al., 2017), and GONE2 (Santiago et al., 2025), and the fitted model was validated by forward simulation. The coalescent class was tested by comparing the folded spectrum with Kingman-plus-growth and Beta-coalescent expectations. Statistics that require individual genotypes were computed on the called genotypes with pg_gpu (Pope et al., 2026). These are the windowed per-arrangement π and d_{xy}, the between-karyotype and between-division Hudson F_{ST} (Bhatia et al., 2013), linkage disequilibrium, and the principal-component analyses, with missing genotypes retained and handled pairwise. Per-SNP geographic F_{ST} was computed with plink2 (Chang et al., 2015) using

the Weir–Cockerham estimator. Relatedness was screened with KING-robust (Manichaikul et al., 2010).

The chromosome-2 inversion. Karyotypes were assigned by local PCA (Li and Ralph, 2019) and a Gaussian-mixture model on the first principal component. The two haplotype assemblies of the reference individual were aligned to each other, to the primary assembly, and to *I. coindetii* with minimap2 (Li, 2018), and the orientation of the inversion relative to *I. coindetii* was confirmed with an AnchorWave (Song et al., 2022) whole-genome alignment. The age and frequency trajectory were fitted with msinv (Small, 2025), a sequentially Markov coalescent simulator that enforces the recombination barrier between arrangements (Supplementary Methods). The fit was a grid search, not approximate Bayesian computation. Each cell of a grid over the barrier age, the plateau frequency, and the timing and duration of the rise was simulated 48 times and scored by a covariance-weighted distance D between the simulated means of π_S/π_I, d_{xy}/π_S and the singleton ratio and their empirical values, with the measurement covariance and the variance of the frequency trajectory under drift as weights, and each parameter was bounded by a ΔD profile with a cutoff calibrated by parametric bootstrap rather than a posterior distribution. Gene content, fixed differences between arrangements, gene-ontology enrichment against a permutation null, the drift bounds, and the mutational-load test are described in the Supplementary Methods.

Recombination. A recombination map was estimated with ReLERNN (Adrion et al., 2020), which trains a recurrent neural network on coalescent simulations matched to the fitted demography and the accessibility mask and then predicts the rate from linkage disequilibrium in windows of 3–6 kb. The map was built from the 329 confidently sexed males that had calls on every chromosome, one of 330 being removed, and a bootstrap correction with 200 replicates was applied. The Z was estimated with a separate network.

Selective sweeps. We classified 100-kb windows with diploS/HIC (Kern and Schrider, 2018), trained on SLiM forward simulations (Haller et al., 2026) recapitated with msprime (Kelleher et al., 2018; Haller et al., 2019) and on neutral msprime simulations under the fitted demography, at the empirical sample size and with the empirical pattern of missing data. Training and scanning used the same unphased feature extractor and the full site-frequency spectrum. Each window's sweep score was ranked within strata of callable fraction and coding density, and outliers were corroborated with RAiSD (Alachiotis and Pavlidis, 2018) and SweepFinder2 (DeGiorgio et al., 2016). A recombination-aware hidden Markov model resolved the window classifications into spatially coherent Tier-1 calls. The Z was scanned in males only with a model retrained at their sample size. The extended mask excluded the 22 anomalous-diversity blocks throughout. Candidate genes were annotated with EnTAP (Hart et al., 2020) and checked against cephalopod proteomes with DIAMOND (Buchfink et al., 2021). Gene-ontology enrichment used Fisher's exact test with Benjamini–Hochberg correction (Benjamini and Hochberg, 1995), and the background-selection screen retained calls whose local diversity lies at least 0.003 below the value expected from coding density within 1 Mb (Supplementary Methods).

McDonald–Kreitman tests. McDonald–Kreitman tests used mkado (Rivera-Colón et al., 2026) with *I. coindetii* as the outgroup for 20,999 genes, with codon degeneracy annotated by degenotate (Thomas, 2024).

Data and code availability

The manuscript, along with all figures and tables, their generating scripts, and editable table exports, is available in the manuscript repository (https://github.com/stsmall/Illex_popgen). A section-by-section index of every figure and table source file is provided in `FIGURES_AND_TABLES_INDEX.md`. The annotation utilized is `Illex_F24` version 3, which includes additions to version 2, specifically the loci recovered in the inversion and the chromosome-16 functional note, located in `data/annotation/` within the same repository. Whole-genome sequencing reads for *I. argentinus* will be deposited in the NCBI Sequence Read Archive and the variant calls (VCF) in figshare [accessions to be added]. The coalescent simulations for the inversion analysis (section 2.2) were generated using `msinv` v0.3.5, a general ARG-based inversion-coalescent simulator (https://github.com/stsmall/msinv, MIT, commit `35412bf`). The complete *Illex illecebrosus*-specific inversion analysis, including fitting code, simulation configuration, tabulated outputs, and inversion figures, is contained within the repository under `illex/` and `results/illex/`.

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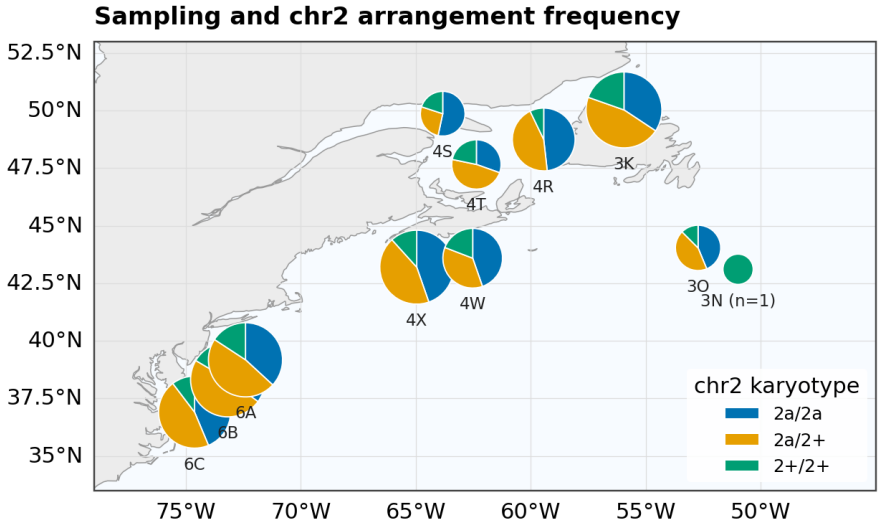


Figure 1: **Sampling and chromosome-2 arrangement frequency.** *Illex illecebrosus* was sampled across eleven NAFO divisions (approximately 37–50°N). The pie charts, positioned at the mean sampling coordinates of each division, illustrate the karyotype composition (2a/2a, 2a/2+ and 2+/2+), with sizes proportional to N. The frequency of the derived (2a) arrangement is non-clinal (ranging from 0.54 to 0.71 across latitudes), which aligns with a geographic $F_{ST} \approx 0$, indicating that the inversion is not locally adaptive.

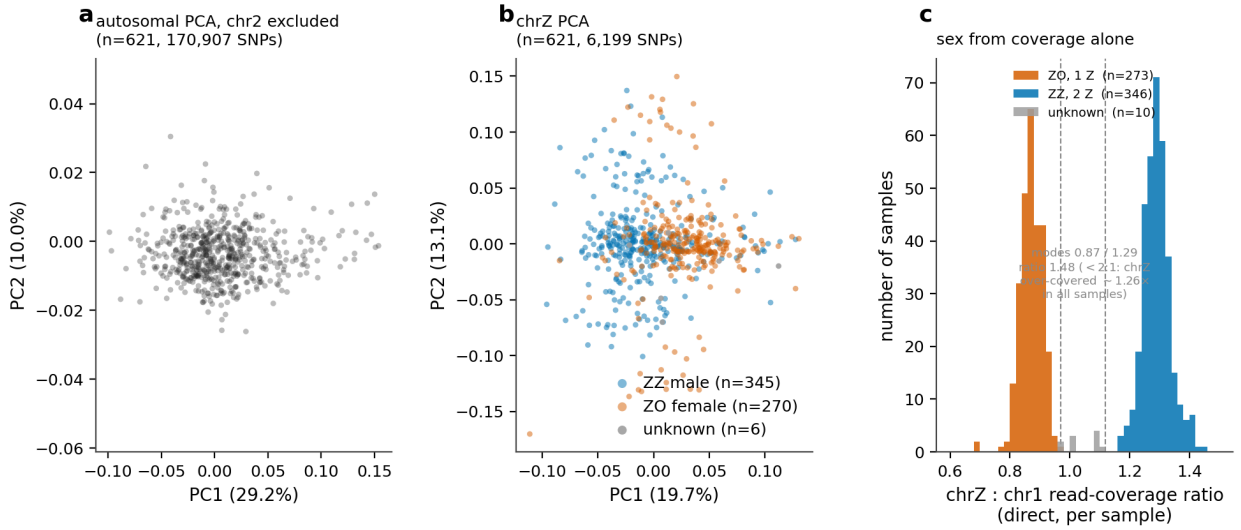


Figure 2: **Absence of population structure outside chromosome 2 and sex assignment** (consistent with the range-wide panmixia reported by Baker et al. 2025). **(a)** Genome-wide autosomal PCA excluding chromosome 2 reveals a single undifferentiated cluster. **(b)** PCA of chrZ shows mild separation between males and females. **(c)** The chrZ:chr1 read-coverage ratio per individual indicates ZZ males at approximately 1.0 (two Z copies), ZO females near the single-copy expectation (absence of W), with individuals in the ambiguous band remaining unassigned. The sample includes 342 males and 291 females.

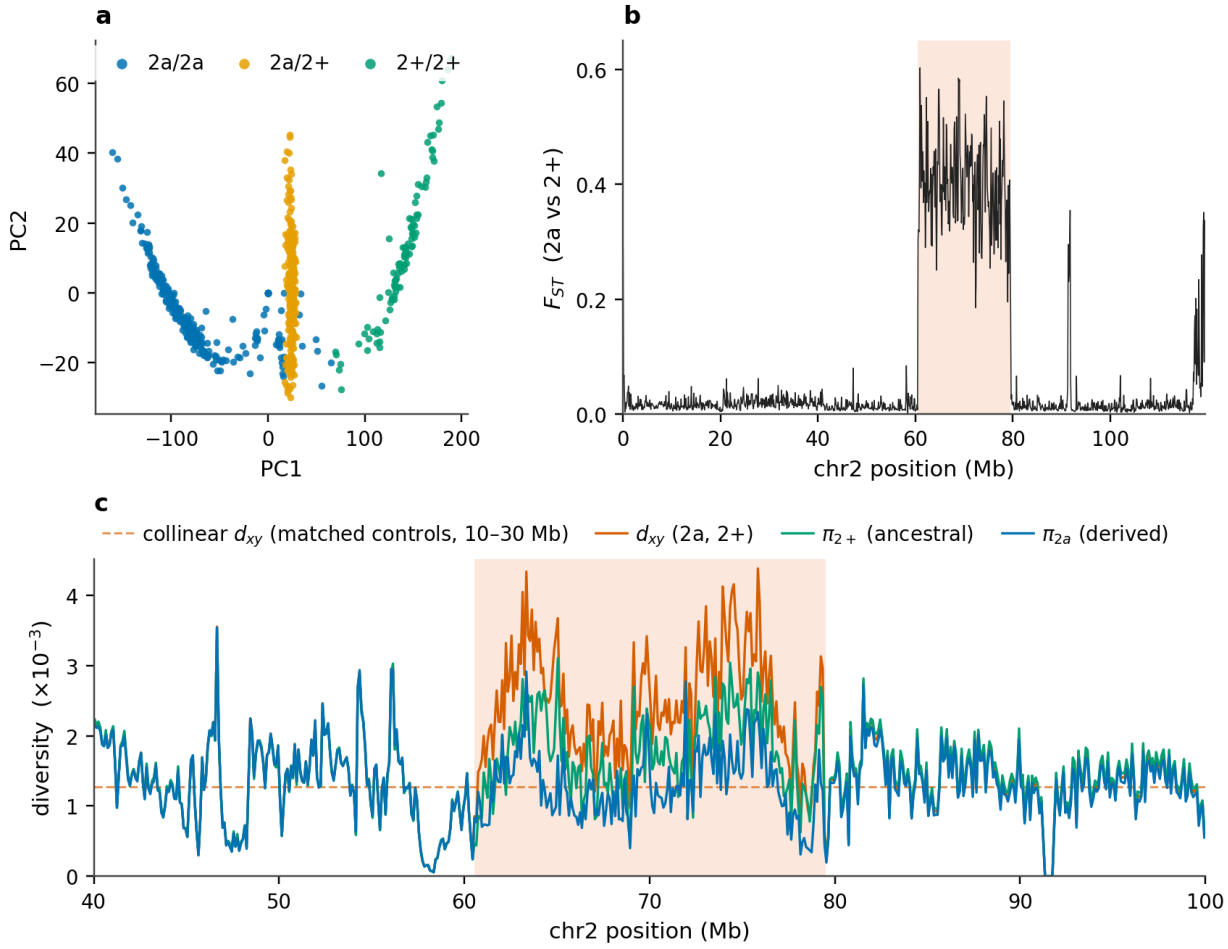


Figure 3: **Chromosome-2 inversion analysis.** **(a)** PCA of chr2 distinguishes three karyotype clusters (2a/2a, 2a/2+, 2+/2+). **(b)** F_{ST} values between arrangements along chr2 indicate that the inversion (60.54–79.50 Mb, shaded) is the sole differentiated region (median $F_{\mathrm{ST}} = 0.37$ compared to 0.01 in collinear regions, with a comprehensive view in Supplementary Fig. S20). **(c)** Diversity per arrangement across the inversion (shaded) and 20 Mb of collinear sequence on either side shows that within the breakpoints, the derived 2a arrangement is less diverse than the ancestral 2+ ($\pi_I/\pi_S = 0.74$), despite its higher frequency, and d_{xy} surpasses both, indicating an ancient polymorphism whose derived arrangement was rarer in the past. Outside the breakpoints, the three curves converge at the collinear d_{xy} (dashed), as expected where arrangements recombine freely.

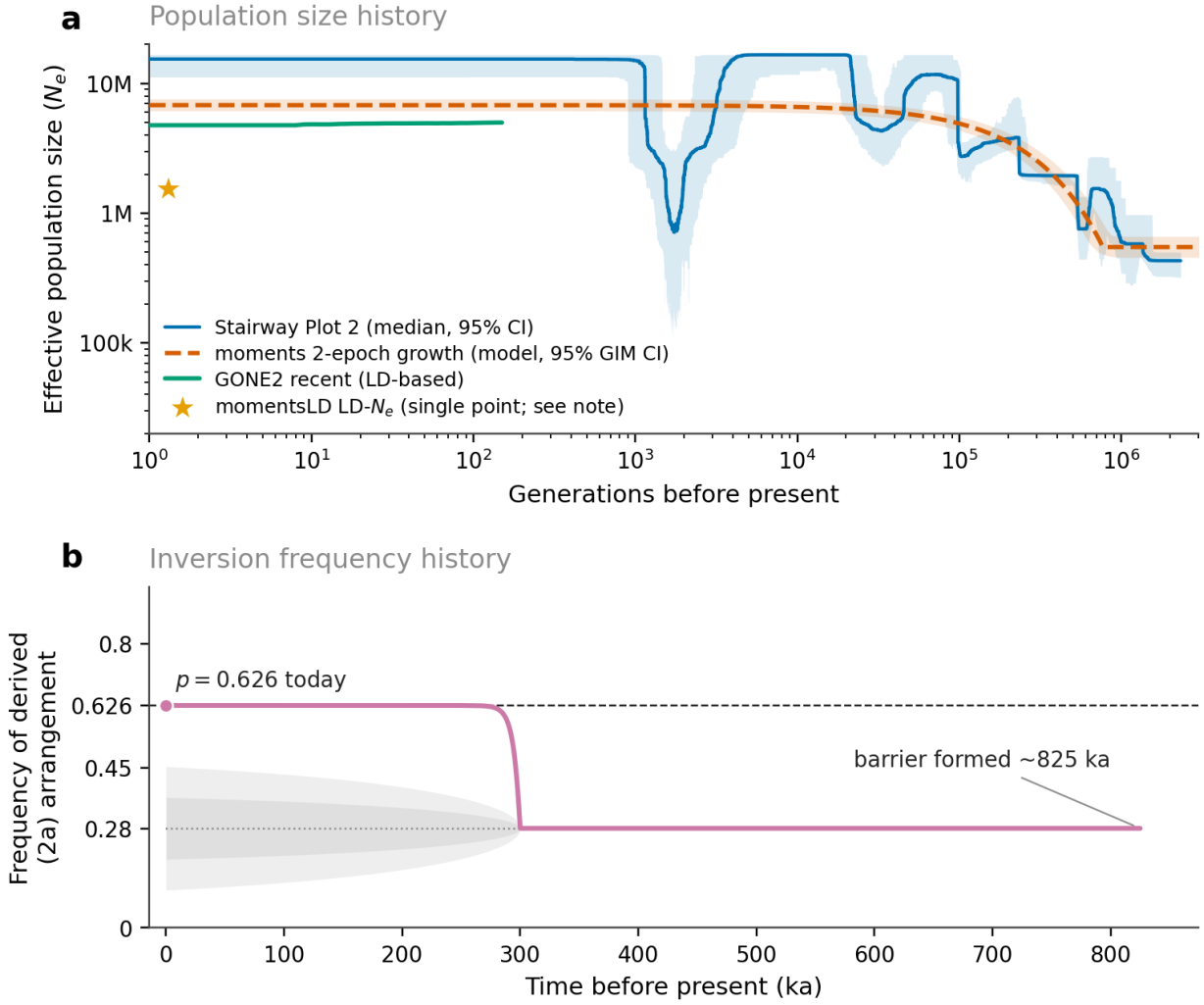


Figure 4: **Demographic history and inversion frequency history. (a)** The effective population size over time is depicted using Stairway Plot 2 (median ±95% CI), alongside the moments 2-epoch growth model with its Godambe (GIM) 95% parameter-CI envelope (expansion $12.4 \times \pm 9\%$, $T \approx 770$ ky $\pm 16\%$). GONE2 is utilized for recent historical data, and a single momentsLD LD-N_e point is presented (rate-corrected, with only the contemporary point shown due to the non-identifiability of its time-varying fit). **(b)** The inversion trajectory is fitted using the barrier-aware coalescent simulator msinv. The recombination barrier formed approximately 800–850 ka, coinciding with the onset of expansion in (a), and the derived (2a) arrangement was held near $p \approx 0.22$–0.38 (best fit 0.28 shown) and subsequently rose to 0.626. The neutral-drift envelope is provided for comparison. ΔD profiles, which identify the end and duration of the rise, are available in Supplementary Fig. S14, indicating that the rise was recent and rapid, excluding neutral drift.

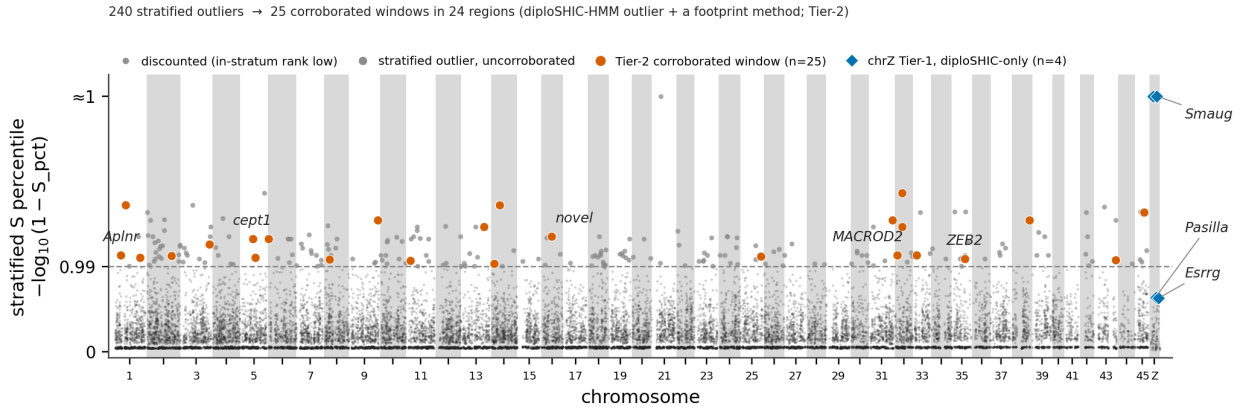


Figure 5: **Genome-wide selective-sweep scan.** The sweep signal across the genome (chromosomes concatenated) is illustrated. The classifier interprets the genome as predominantly linked-selected (∼91% linked-hard or linked-soft, ∼1% neutral, and only a few percent hard or soft). Raw diploS/HIC calls are filtered by cross-method concordance (a diploS/HIC-HMM outlier plus RAiSD or SweepFinder2) and a background-selection test. The corroborated candidates, robust against background selection (25 windows across 24 regions, highlighted), are distinguished above the discounted background, with key genes labeled. Chromosome-Z Tier-1 calls (diamonds, with *Pasilla*, *Smaug*, *Esrrg* labeled) are male-based and diploS/HIC-only (the footprint methods do not cover the Z), thus they cannot be included in the concordance filter. The chromosome-2 inversion does not exhibit a concordant sweep. The complete scan pipeline is detailed in Supplementary Fig. S8.

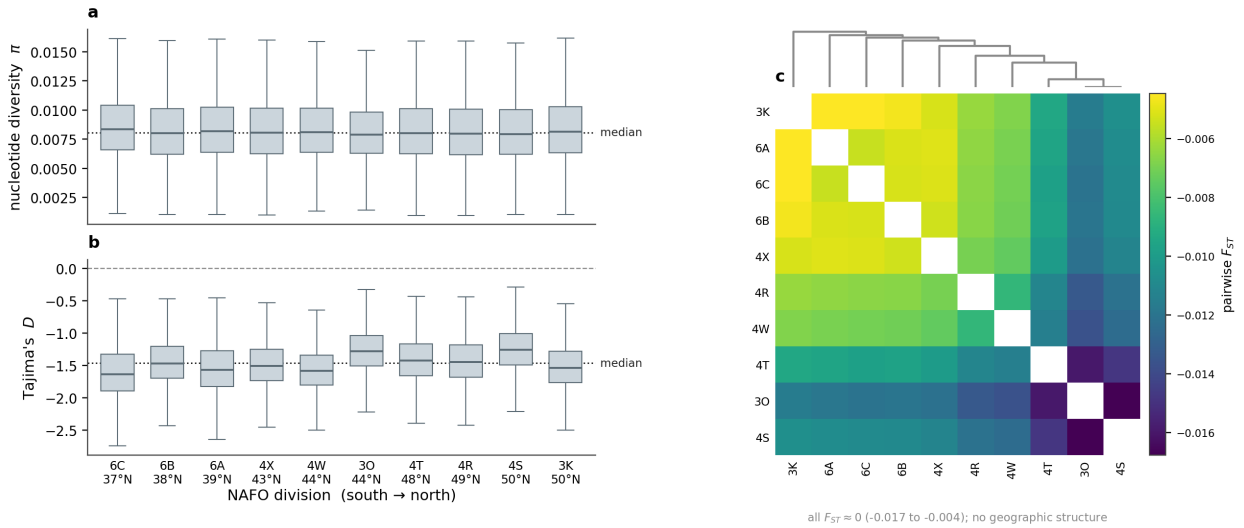


Figure 6: **Diversity is uniform and there is no geographic structure.** **(a)** Nucleotide diversity π and **(b)** Tajima's D per NAFO division (ten of the eleven sampled divisions, as 3N, with a single individual, cannot support per-population estimates), ordered from south to north, and both are flat across latitude (no cline), and D is negative in every population, the signature of strong expansion. **(c)** Pairwise F_{ST} among divisions as a clustered heatmap with dendrogram. Every value is ≈ 0 (-0.017 to -0.004) and the clustering follows sample size, not geography, i.e. no population structure. (π/D estimated by ANGSD genotype likelihoods over all 45 autosomes plus the Z, large divisions subsampled to $n = 25$, and F_{ST} genome-wide by pg_gpu. Per-chromosome distributions are in Supplementary Fig. S9.)

Table 1: **Sample and whole-genome summary.** Diversity statistics are genome-wide from genotype likelihoods, and karyotype counts and the derived-arrangement frequency are for chromosome 2.

Quantity	Value
Species / callset	*Illex illecebrosus*; Baker et al. (2025) whole-genome
Individuals (karyotyped)	633
Sex (chrZ coverage)	342 ZZ males / 291 ZO females
Sampling	11 NAFO divisions, ∼37–50°N
Geographic structure	$F_{ST} \approx 0$ (none outside chr2)
Genome-wide π	0.0093
Genome-wide Tajima's D	-2.07 (-2.09 in 50 kb windows)
chr2 karyotypes (2a/2a, 2a/2+, 2+/2+)	254 / 284 / 95
Derived (2a) frequency	0.626 (0.54–0.71 across divisions, non-clinal)
Karyotype F_{ST} (2a/2a vs 2+/2+)	0.365

Table 2: **The chromosome-2 inversion** (I = 2a derived and S = 2+ ancestral).

Quantity	Value
Location	chr2:60.54–79.50 Mb (∼18.96 Mb)
Karyotypes (2a/2a, 2a/2+, 2+/2+)	254 / 284 / 95 (95.3% confident)
Derived (2a) frequency p	0.626
Historical plateau frequency	0.22–0.38
Karyotype F_{ST} (2a/2a vs 2+/2+)	0.365 (geographic $F_{ST} \approx 0$)
π_I/π_S (I=2a)	0.737 ± 0.026
d_{xy}/π_S	1.385 ± 0.021
Coalescent age t_{inv}	∼800–850 ky ($\propto 1/\mu$, $\mu = 3 \times 10^{-9}$)
Rise	0.22–0.38 → 0.626 over 50–200 ky, ending 100–250 ka
Selection on rise	$N_e s \sim 10^2$–10^3 ($s \approx 1.2$–5×10^{-4})
Gene content	89 gene models, 45 annotated; no GO enrichment ($p = 0.51$)
Fixed differences (2a vs 2+)	3,513; 19 exonic, in 9 genes

Table 3: **Demographic model** (moments, folded SFS, validated by forward simulation).

Parameter	Value
Ancestral N_e (N_REF)	547,928
Present N_e (N0)	6,808,096
Growth duration (T_GROW)	769,519 generations
Fold expansion	∼12×
Genome-wide π	0.0093
Genome-wide Tajima's D	≈ -2 (nearly all windows)
Mutation rate μ	3×10^{-9} / bp / generation
Recombination (autosomal)	∼0.24 cM/Mb (sex-averaged)

Table 4: **High-confidence selective-sweep candidates**, grouped by biological function and ordered by chromosome, including the Z-chromosome set. Autosomal rows are the 11 background-selection-robust candidates (of 24 concordant regions), and Z rows (marked) are the fine-scale-confirmed sweeps from the males-only scan. Genes are cephalopod orthologs. Class is hard or soft sweep. Full machine-readable tables are in Supplementary Table S6.

Location	Gene / feature	Class
Neural, RNA-regulation & development		
chr35:39.0 Mb	*ZEB2* (neural/EMT transcription factor)	hard
Z:20.8 Mb	*Pasilla*/Nova-1 (splicing regulator)	hard (Z)
Z:23.6 Mb	*Smaug*/samd4a (translational repressor)	hard (Z)
Cardiovascular		
chr1:20.3 Mb	*Aplnr* (apelin receptor)	hard
Genome-maintenance & stress		
chr32:9.3 Mb	*MACROD2* (DNA-damage response)	soft
Membrane & lipid		
chr5:38.0 Mb	*cept1* (phospholipid synthesis)	hard
chr1:85.5 Mb	*Cpes* (ceramide phosphoethanolamine synthase) †	hard
Metabolism		
chr25:65.0 Mb	*GALT* †	soft
ECM & adhesion		
chr33:9.0 Mb	*Tinagl1*	hard
Signalling		
chr32:24.1 Mb	GPCR cluster	hard
Z:30.7 Mb	*Esrrg* (estrogen-related receptor)	soft (Z)
Cephalopod-specific novelty		
chr16:35.3 Mb	uncharacterised (*Euprymna* family)	hard
Lower confidence (TE-dense)		
chr1:35.9 Mb	transposable element ‡	hard
chr11:10.8 Mb	transposable element ‡	hard

† target reassigned by fine-scale localisation to a neighbouring annotated gene. ‡ TE-dense region, where deep π and negative D may be a TE-insertion sweep or a repeat artifact.